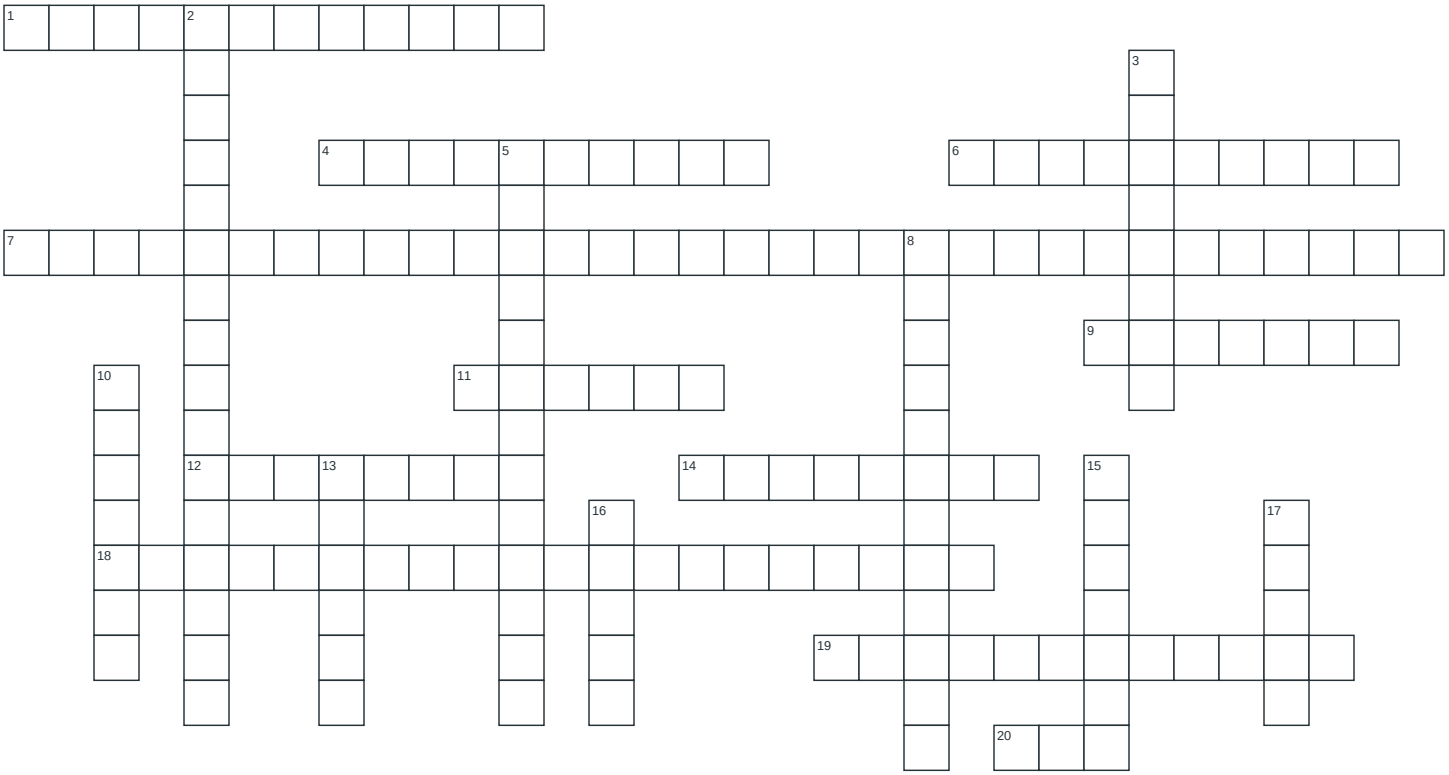
Periodic Table and Trends Crossword

Hard | 20 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

1. Number of protons in an atom’s nucleus
4. Group that includes helium, neon, and argon
6. Have some properties associated with each group
7. Holds oppositely charged ions together in an ionic solid
9. Pure substance made of only one type of atom.
11. Subatomic particle that carries a positive electric charge
12. Definition of a beta-minus particle emitted in nuclear decay
14. Group that includes fluorine, chlorine, and bromine
18. Condition in which positive and negative charges balance.
19. Increases as additional occupied electron shells are added
20. Atom or group of atoms with a net electric charge.

DOWN

2. Energy required to remove an electron from a gaseous atom or ion
3. A discrete group of atoms held together by covalent bonds
5. Electron locations are described by regions of probability
8. Sharing of electron pairs between atoms
10. Membrane-bound organelle that contains most eukaryotic DNA.
13. Element that forms the backbone of most organic molecules
15. Subatomic particle that has no electric charge
16. An unfamiliar element readily conducts electricity and can be hammered into a sheet. Which classification best fits
17. Definition of a vertical column of the periodic table called

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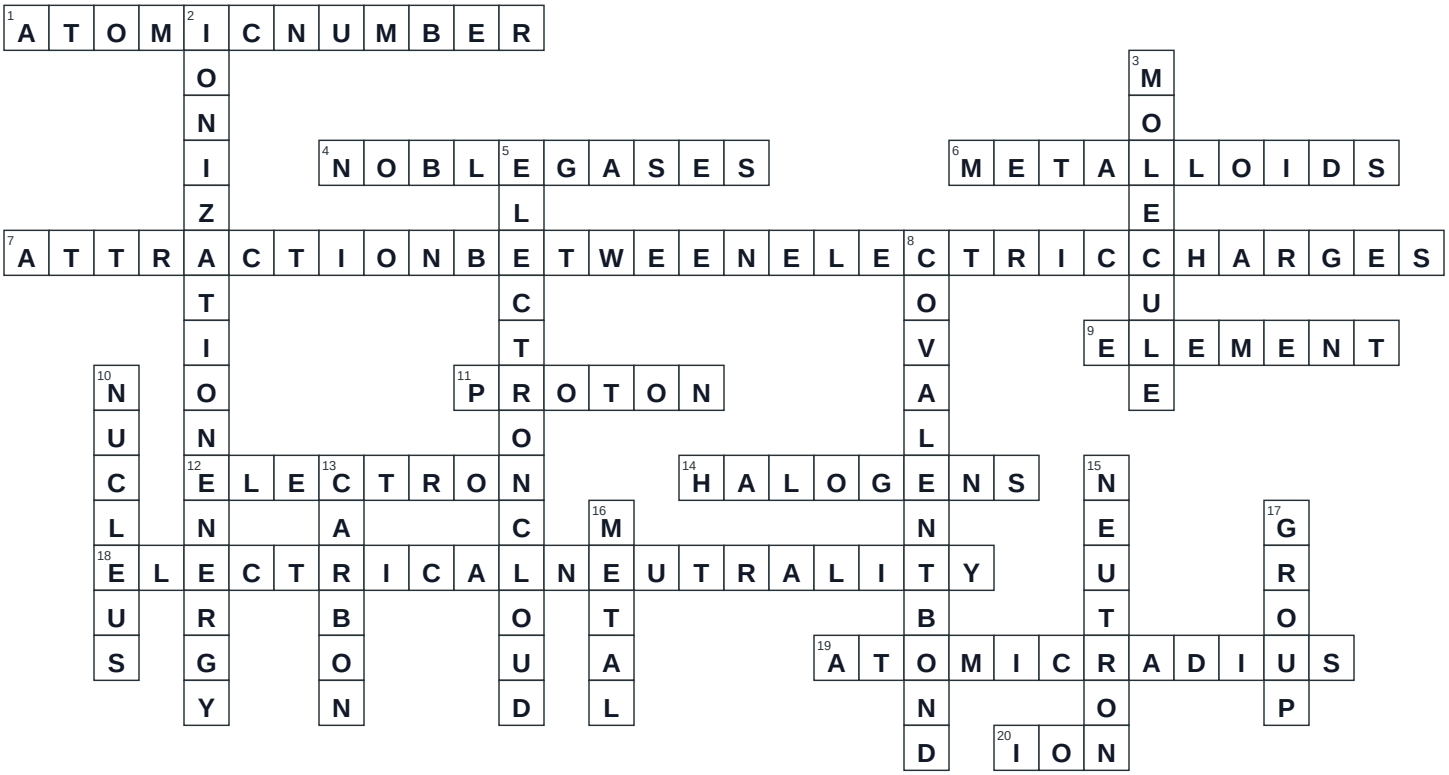
Hard | 20 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

1. Atomic Number
4. Noble Gases
6. Metalloids
7. Attraction Between Electric Charges
9. Element
11. Proton
12. Electron
14. Halogens
18. Electrical neutrality
19. Atomic Radius
20. Ion

DOWN

2. Ionization Energy
3. Molecule
5. Electron Cloud
8. Covalent Bond
10. Nucleus
13. Carbon
15. Neutron
16. Metal
17. Group

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